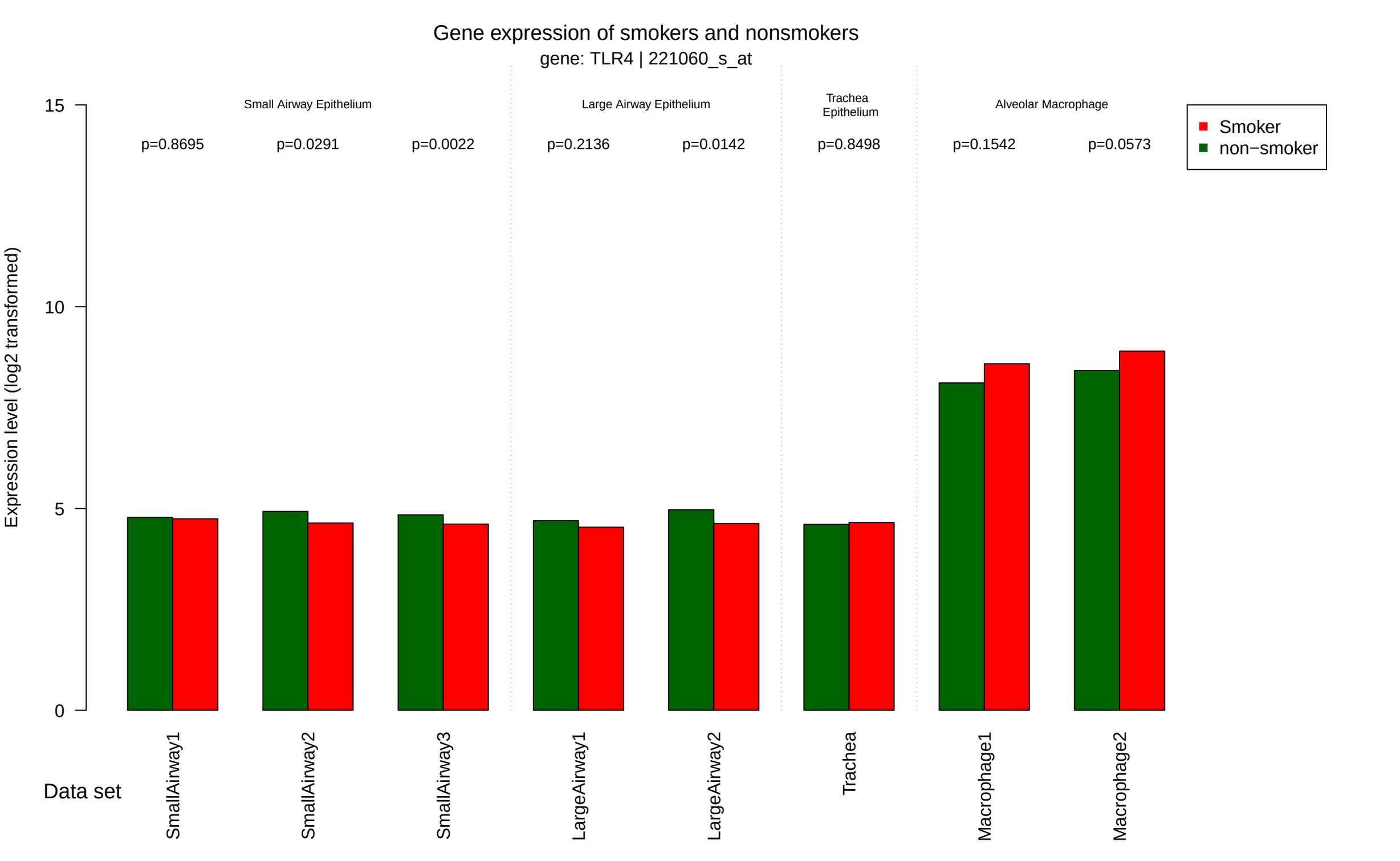
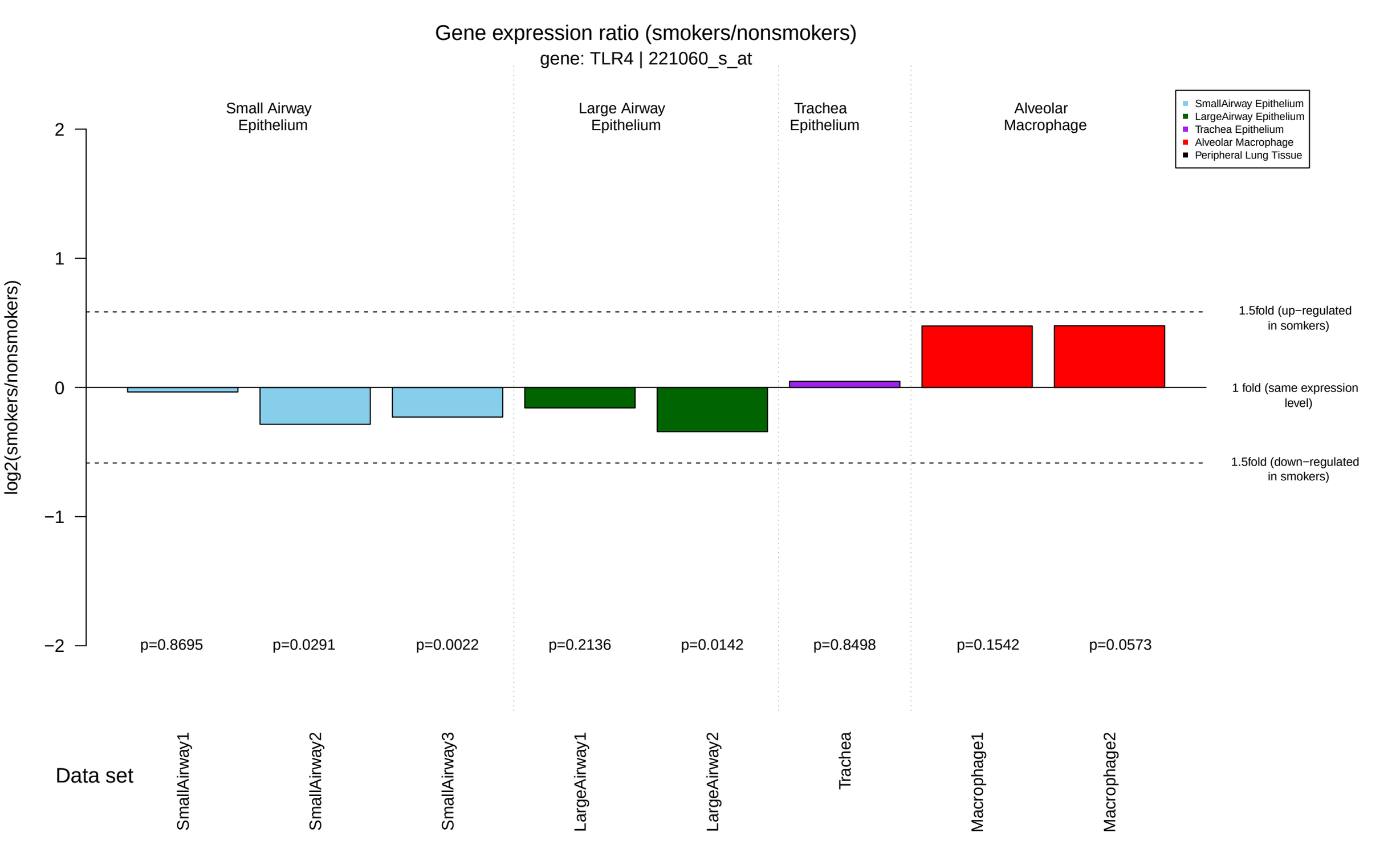
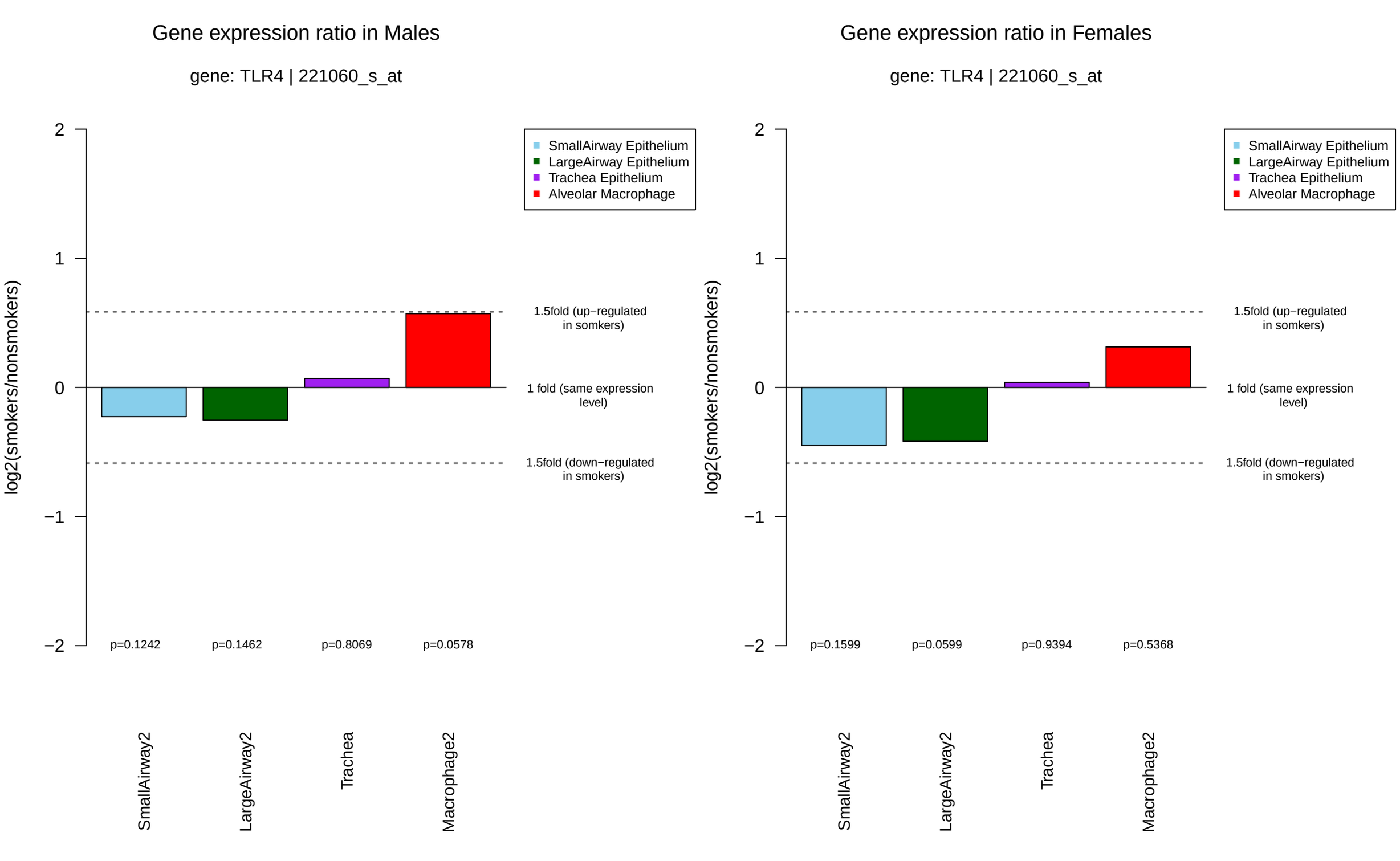
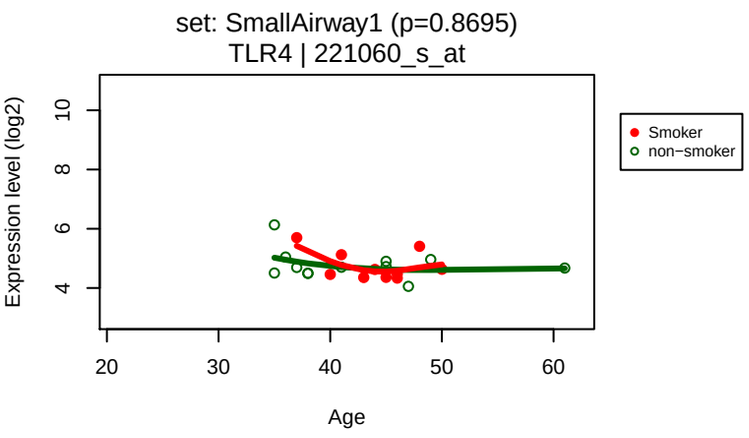
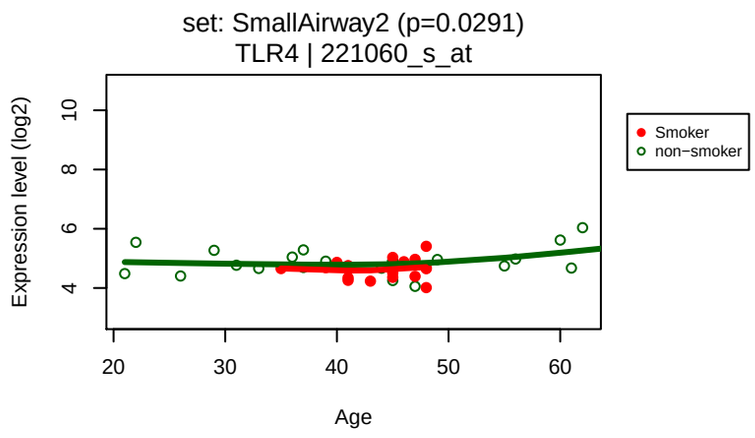
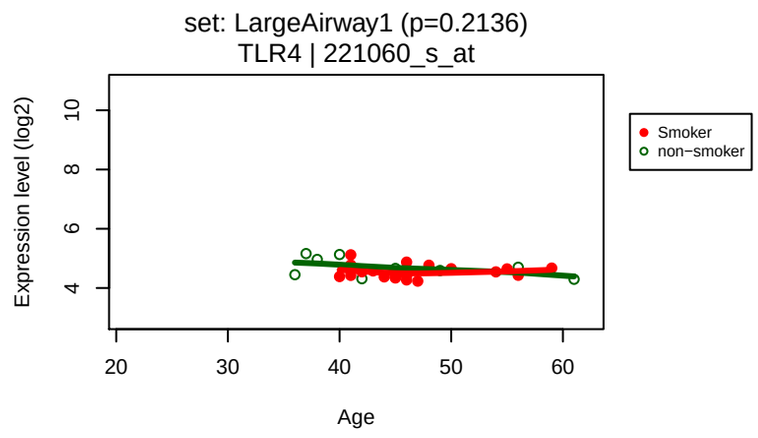

Gene expression of smokers and nonsmokers
gene: TLR4 | 221060_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8695
p=0.0291
p=0.0022
p=0.2136
p=0.0142
p=0.8498
p=0.1542
p=0.0573
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

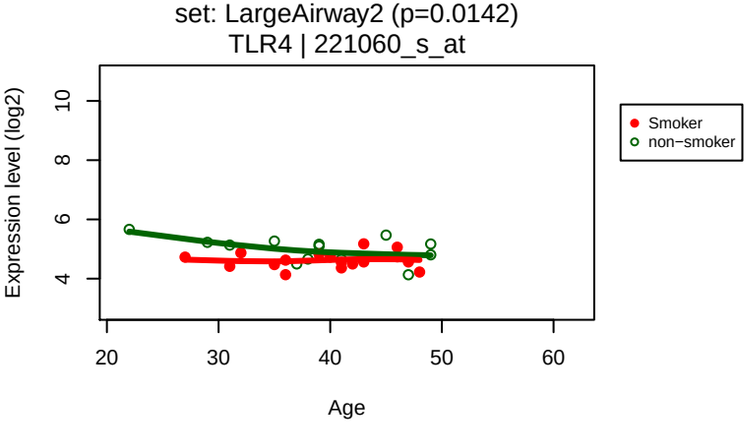

Gene expression ratio (smokers/nonsmokers)
gene: TLR4 | 221060_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8695
p=0.0291
p=0.0022
p=0.2136
p=0.0142
p=0.8498
p=0.1542
p=0.0573
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

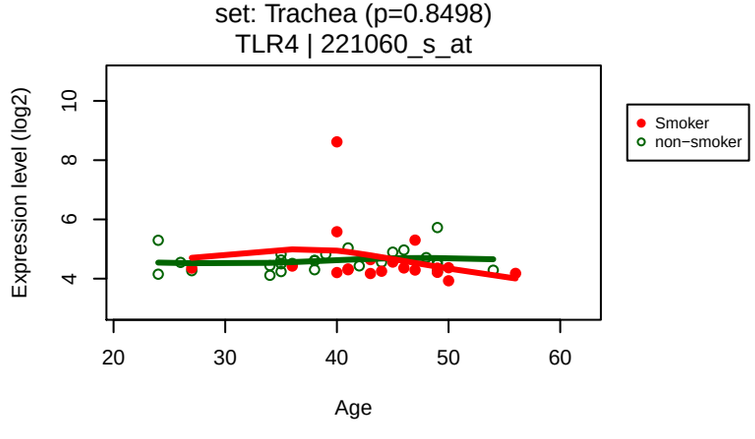

Gene expression ratio in Males
gene: TLR4 | 221060_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1242
p=0.1462
p=0.8069
p=0.0578
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TLR4 | 221060_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1599
p=0.0599
p=0.9394
p=0.5368
SmallAirway2
LargeAirway2
Trachea
Macrophage2

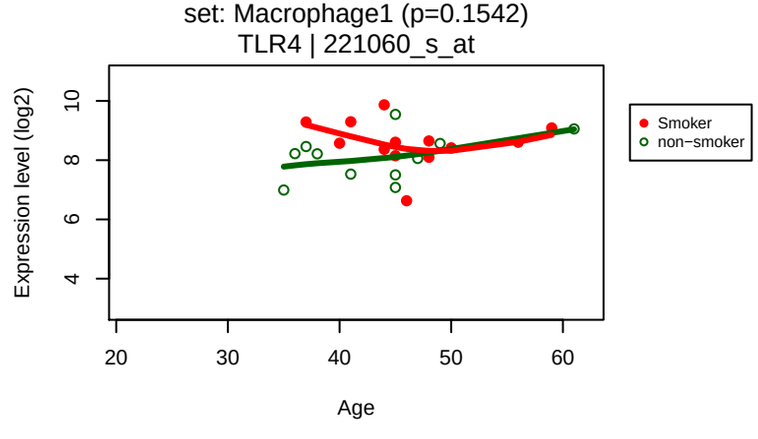

set: SmallAirway1 (p=0.8695)
TLR4 | 221060_s_at
Expression level (log2)
Age
Smoker
non−smoker

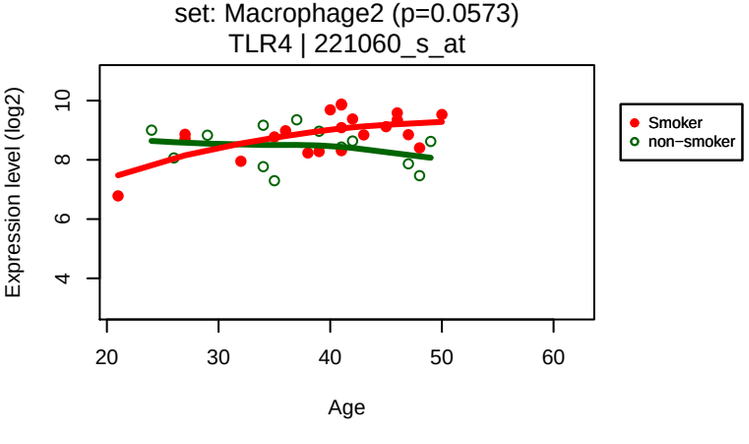

set: SmallAirway2 (p=0.0291)
TLR4 | 221060_s_at
Expression level (log2)
Age
Smoker
non−smoker

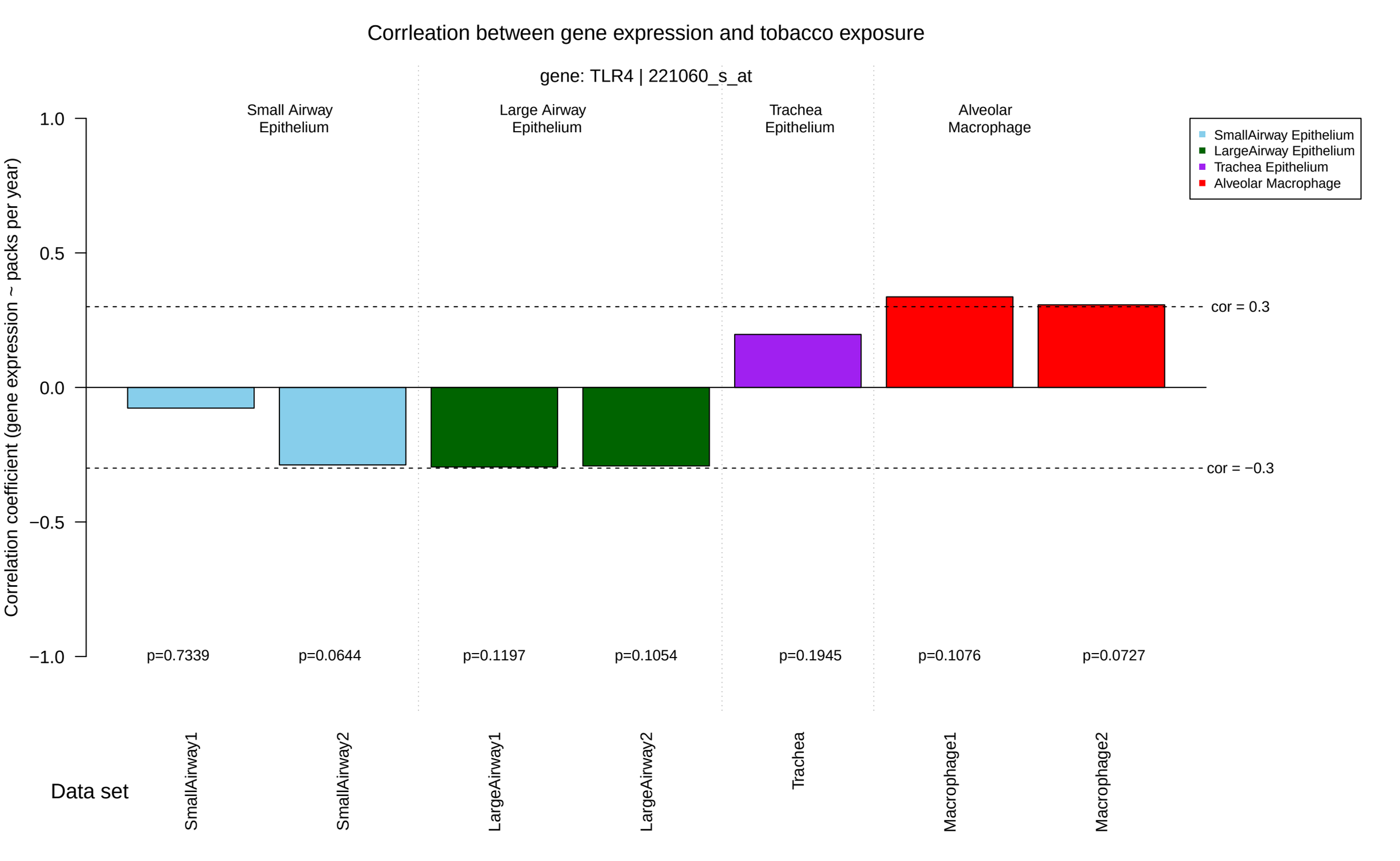

set: LargeAirway1 (p=0.2136)
TLR4 | 221060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0142)
TLR4 | 221060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8498)
TLR4 | 221060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1542)
TLR4 | 221060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0573)
TLR4 | 221060_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TLR4 | 221060_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7339
p=0.0644
p=0.1197
p=0.1054
p=0.1945
p=0.1076
p=0.0727
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2